The Whole Thing, In Plain Words

No jargon, plus a glossary of every term used elsewhere

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The whole thing, in plain words

There is no jargon in this document. Every technical term used in the other documents is defined at the end.

The situation

A bank uses a computer program to decide who gets a mortgage. The program looks at income, loan size, property value and so on, and it says yes or no.

Someone checks the results and finds a problem: the program approves **79%** of one group of applicants but only **57%** of another, even though the applications are the same kind. This is what people call "algorithmic bias", and there are well-known tools for fixing it. You tell the program to approve both groups at the same rate.

The catch, and it is a known one

There are two completely different ways for the program to obey that instruction.

Way one: approve more people from the group that was being turned down.

Way two: turn down more people from the group that was getting through.

Both ways make the rates equal, and both count as "fixed". The problem is that the score which certifies the fix cannot tell them apart, because it only measures whether the two rates match, not how they came to match.

Way one helps people. Way two helps nobody, because it just spreads the disadvantage around. The ethics literature has a name for it: *levelling down.*

What was already known

A well-known 2023 paper from Oxford showed that these tools mostly do way two, and they put it in their title: levelling down happens "by default".

The fact that the score cannot tell way one from way two was also known. A 2023 paper on fairness across combinations of groups says it in plain words, that this "often goes unnoticed in the overall performance of the model", and a separate 2023 auditing tool exists precisely because a fairness score does not say how many people were affected or in which direction. This means the catch above is where this work starts, not something it found.

The Oxford paper is also the closest competitor, because it not only identified the problem but proposed the fix. This project originally believed it had invented that fix independently. It had not, which was found by reading their paper properly, and corrected.

The way of measuring the effect is not mine either. A 2023 paper set out how to audit what one of these fixes does to individuals: how many people it moves, which way it moves them, and what happens to the overall approval rate. Those are the three things counted here. What is mine is what the counts show once you run them on 26 runs across 15 populations instead of one, which is that the direction depends on the data. Their framework was never used to ask that question.

There is a second piece of prior work. A March 2026 paper proves mathematically that when the program cannot see which group someone belongs to, which is the situation studied here, the effect **can go either way** depending on the data. So the idea that it is not always harmful was published, in theory, five months before this work reached it independently. That paper has no experiments in it at all, not one dataset. It proves the possibility and gives conditions written in terms of the shape of the program's internal scores.

What this work found

It is not "by default". It depends on how generous the system is to begin with.

Count what fraction of everyone gets approved, across both groups, and call that the approval rate of the system as a whole.

- If the system approves fewer than about half of everyone, forcing equality takes approvals **away**. That is way two, the harmful one.
- If it approves more than about half, forcing equality **hands approvals out**. That is way one, the good one.
- The switch sits close to an approval rate of 0.54, and everywhere it has been measured carefully it lands between roughly 0.43 and 0.58.

This means the same tool, given the same instruction, has opposite effects on real people, and what decides between them is nothing more than how generous the system already was.

How it was proven, and why the proof is trustworthy

The obvious worry is that two datasets differ in a hundred ways, so you can never tell which difference caused what. The test therefore used one dataset and changed exactly one thing.

The dataset predicts whether a person earns more than \{}$50,000, and that \{}$50,000 is an arbitrary number someone picked. Move it to \{}$70,000 and fewer people qualify, while moving it to \{}$20,000 means most people do. Same people, same information about them, same groups. Only the finish line moved, and the effect flipped exactly as described. There is automated test code that checks the people, the information and the groups really are identical between runs, so this is not a matter of trust.

It then held up in a second population, in a second way of dividing people into groups, in real mortgage data where there is no arbitrary cutoff at all because the answer is a real lender's real decision, and in fourteen further runs on four new populations carried out after the prediction was written down, which means they could not have influenced it. Four is the number that matters there, because runs on the same population share their people and differ only in where the cutoff was put.

What happened when I tried hard to break it

I attacked this finding four ways, on five populations each instead of one. Two of the attacks it survived, while two found real limits, and both limits are reported here rather than tucked away.

I tried	Asking	What happened
a completely different kind of model	is this just true of simple models?	**held, 5 out of 5**
making the fairness rule much stricter, and much looser	is it just one setting?	**held, 4 out of 5**
reaching the same approval rate a different way	is it the approval rate, or the difficulty?	**held, 4 out of 5**
a different definition of fairness	does it hold for any fairness rule?	**it failed**

The third one matters most. The original evidence changed the approval rate by changing what counted as success, earning over \{}\$20,000 instead of over \{}\$100,000, but those are not the same task, because one is much harder to predict than the other. So the old evidence could never separate "the approval rate is what drives this" from "the difficulty is what drives this". The new test changes nothing about the task at all: same people, same information, same predictions, and only the cut-off the model uses to say yes moves. The effect reproduced, which shows the approval rate is what matters. It also cleared up a puzzle I could not explain before, because the switch point sits in a different place for mortgages than for income, and that turns out to be a real difference between the two worlds rather than a mistake in how I built the earlier evidence.

The fourth attack failed, and I am not going to dress it up. There is a second common definition of fairness, concerned with error rates rather than with how many people get approved. Under that definition the relationship almost disappears, and adding more populations made it weaker rather than stronger. So the finding is narrower than I first claimed: it is about fairness rules that control how many people are approved, not about fairness rules generally.

One population gave no answer at all. In Connecticut almost everybody is above the switch point already, so the fairness rule barely changes anything and there is nothing to measure. That turned up a flaw in my own method, because my original test demanded that the relationship be strong but never demanded that anything actually move. I added that requirement and re-checked every earlier result against it. Three go from "failed" to "no answer", and none of them had ever passed by accident, so nothing I claimed turns out to be wrong. But the hole was mine, and it was found by a population rather than by me.

Two things I learned since the last version

The first is that where the switch sits is much more predictable than I thought. Measuring it properly, with more points and after throwing away any test where the model had become worse than useless, four completely different populations agree:

population	field	switch point
COMPAS	criminal justice	0.511
South Carolina	income	0.530
Oregon	income	0.558
Dutch census 2001	jobs	0.576

Three fields, two countries and four separate data sources all land between 0.51 and 0.58, which means the advice is now "expect about 0.54 and check" instead of "you will have to work yours out from scratch".

The second is that the finding belongs to one specific situation, which the theory predicted. There are two ways to make a system fairer. You can change how it is trained, which is what I do, or you can leave the training alone and adjust the decisions afterwards. The second way requires knowing each person's race or sex at the moment you decide, and that is illegal in most places for exactly that reason.

I tried the second way across seventeen populations, and my finding disappeared completely. That sounds bad, but it is the opposite. The theory paper says these two situations are fundamentally different: when the program can see the protected attribute, the outcome is guaranteed, with the disadvantaged group always gaining and the advantaged group always losing, while when it cannot, the outcome could go either way, which is exactly when a rule for telling which is worth having. So there was nothing for my rule to predict in the second case, and the theory's prediction for that case held in **27 out of 27** of my populations with no exceptions, nine of those predicted in advance on populations that had never been measured that way before.

My claim is now smaller and firmer. It is about systems that cannot see the protected attribute when they decide, which is most of them, because looking at race or sex at the moment of the decision is illegal in most regulated settings.

How this sits with the theory

Three things came out of checking this work against that 2026 paper.

1. **Their conditions never actually hold.** Applied to all 26 real runs here, the mathematical conditions in their theorem are satisfied zero times, because the quantity they are written over blows up on real data. So their result, as stated, cannot be used on anything.
2. **Their direction is still right.** Loosened into a comparison rather than a strict requirement, it predicts correctly in 24 of 26 cases.
3. **And it matches the approval-rate rule almost exactly.** The two agree on 25 of 26 datasets, and the numbers behind them correlate at 0.93.

So their maths explains why it happens, while this work shows how you can tell in advance, using a number any organisation already has rather than one requiring the model's internals. Two independent routes arrive at the same place, which is better evidence than either alone.

The strongest test: predictions written down before the data existed

Everything above was measured by running experiments and then looking at the results, and the honest problem with that is well known: with enough looking, a pattern can always be found. So the rule was put through the one kind of test that looking cannot pass.

The idea is simple. Write the rule down. Name the datasets it will be tested on, ones nobody has ever measured. Write down the score it must reach, and write down what a lucky guesser would score, because the rule has to beat luck rather than just do well. Save all of that in the project's history, where the timestamps cannot be faked, and only then run the experiments.

The first attempt failed, and the failure taught the most important lesson in the project. Two hours before the test was locked in, I had added a clever extra clause to the rule, based on four datasets I had already seen. With that clause the sealed test scored 4 out of 8, which is no better than guessing, while without it the same test would have scored 7 out of 8. A refinement that looked completely convincing on the data I had was exactly what ruined the prediction on data I had not seen.

So the clause was thrown away, and the plain rule, below the switch point means take away while above means hand out, was sealed against ten more states nobody had touched, with the pass mark set at 9 out of 10 plus the requirement to beat the best possible lucky guesser, which scored 6.

It scored 9 out of 10. The one miss is counted as a miss with no excuses, though it is worth knowing that the missed state's true effect was about four hundredths of a percent and flipped sign between random re-runs, which means there may have been nothing there to predict in either direction.

One more thing matters for reading this project's record. Eleven of these written-down-in-advance tests were run in total, and eight failed. That is not a bad score, because they were not seven attempts at the same thing. Each failure tested a bigger claim than the one that survives, and each one drew a boundary: the rule does not extend to a second definition of fairness, it does not survive into the see-the-attribute world, and it does not come with an explanation of why it works. The five failures are the map of where the claim ends, while the two passes show that inside that map the prediction works. Every failure is printed in the paper, uncorrected, and that is what makes the passes worth believing.

Why it matters

Loans, job applications and university admissions are the places these tools actually get used, and I checked where each one sits.

what is being decided	share who get a "yes"	which way the fix goes
Sifting job applications	**2–3%**	takes opportunities away
Getting into a top university	4–10%	takes opportunities away
Getting into university generally	66–73%	hands opportunities out
Getting a mortgage	**~84%**	hands opportunities out

They sit at opposite ends, which means two organisations can use exactly the same fairness tool, in good faith, and get opposite effects on how many people get helped. The fairness report looking identical either way is not my observation, since it is why Maheshwari et al. and Ferry et al. wrote their papers. What is new is that the direction differs between them, and that you can tell in advance which one you are about to get.

A bank fixing its mortgage model will end up approving more people, while a company fixing its CV-screening will end up interviewing fewer. Same tool, same score, opposite result. That is why knowing which side you are on matters, and why it is worth being able to check.

What could not be worked out

Why the flip happens.

There was an attempt: derive it from theory, write the prediction down in advance, then test it on new data. It passed the test that was set for it, and then a much stupider rule, "more than half get approved? then it is the good kind", did better. So the honest position is that we know when it flips but we do not know why, and that failure is written into the paper rather than left out.

The one number that is worth remembering, with its health warning

About 0.54, call it roughly half. Below that approval rate these fairness tools usually take opportunities away, while above it they usually hand them out, and you can check which side you are on before building anything, using a number you already have.

The health warning comes from our own latest measurements, and it matters. When ten more states were tested, the switch point turned out to sit at 0.28 in Florida and at 0.65 in Pennsylvania, and a few of the largest states had no clean switch point at all. So "roughly half" is the right starting guess and it called 9 out of 10 sealed predictions correctly, but it is a starting guess to check against your own system, not a law of nature. An earlier version of this document said 0.3; better measurement moved it to 0.54, and more measurement has now widened it into a range. Both corrections are part of the record rather than papered over, which is how this project treats every number.

Glossary

Terms used in the other documents in this folder.

Algorithmic bias. A program producing systematically different outcomes for different groups of people, usually because it learned from data in which that difference already existed.

Protected attribute. The characteristic the fairness rule is about, such as sex or race. In this work the programs are never allowed to see it directly.

Proxy. A piece of information that stands in for the protected attribute without being it. If almost everybody labelled "husband" is male, then that label is a proxy for sex, and removing "sex" from the data does not remove the information.

Demographic parity. The rule "approve both groups at the same rate", which is the main fairness rule studied here.

Constraint. The instruction given to the program. "Approve both groups at the same rate" is a constraint.

Mitigation. Any method for reducing bias. This work studies methods that change the program's instructions rather than editing the data.

Selection rate. The fraction of everyone who gets the good outcome. If a bank approves 8,000 of 10,000 applications, the selection rate is 0.8. This is the central quantity in this work.

Base rate. How often the good outcome genuinely occurs in a group, as opposed to how often the program predicts it.

Levelling down. Making two groups equal by making the better-off group worse rather than the worse-off group better.

Levelling up. The opposite, and the desirable version.

"The pie." Informal shorthand for the total number of approvals. Levelling down shrinks the pie, while levelling up grows it.

Exchange rate. How many approvals were destroyed for each one created. Below 1.0 means more were created than destroyed, which is good. On one dataset here it reached 46.7, meaning 46 people lost an approval for every one who gained one.

Baseline. The ordinary program, before any fairness fix, used as the comparison point.

Seed. A random number that decides how data is split for testing. Everything here is run five times with five different seeds, so that a result cannot be an accident of one split.

Held out. Data deliberately not looked at until after a prediction is written down, so the prediction cannot have been shaped by it.

Pre-registration. Writing down what you expect to find, and what would count as being wrong, before running the experiment. The commit history proves the order.

Sealed test. A prediction written down and saved in the project's history before the data it predicts about existed, together with its pass mark and the lucky-guesser score it must beat. This is the strongest form of evidence in the project, and the only kind that cannot be produced by hunting through results until something looks good.

Feature attribution / SHAP. A tool that claims to show which pieces of information a program relied on. This work finds it can move dramatically without the program's actual behaviour changing much.

Intersectional. Looking at combinations, not just "women" and "Black applicants" but "Black women" specifically. A fix can look fine for each group separately while a combination is treated badly.

Correlation (r). A number from -1 to $+1$ for how strongly two things move together. $+0.9$ is a strong positive relationship, while 0 is none.

Partial correlation. The same, but with a third factor held fixed, to check the relationship is not really caused by that third thing.

Adult / ACS / HMDA. Three of the data sources. *Adult* is a 1994 US census extract and the standard benchmark, *ACS* is its modern replacement available per state, and *HMDA* is a public record of real US mortgage applications with the lender's actual decision.

Population. One dataset, such as a state, a cohort or a census, counted once no matter how many experiments run on it. Fifty-seven have been measured.